

PalmMind AI — Junior AI Engineer Hiring Track

Course Outline & Assessment Syllabus

To ensure all candidates possess robust systems-level and AI-engineering competencies, our technical screening tracks evaluate candidates across multiple specific domains.

Phase 1: Local RAG Pipeline Engineering

Candidates must construct a fully functional RAG system in memory or using local database containers. Key skills evaluated include document text extraction, recursive character splitting, semantic sentence boundaries parsing, dense vector generation using local embeddings models (e.g. FastEmbed), and high-efficiency upsert and search query execution in a vector database like Qdrant.

Phase 2: Stateful LLM Agentic Loops

Developing AI agent logic using LangGraph to invoke tool calling, handle database transactions, and persist conversation checkpoints using Redis. The system must operate fully locally using unsloth Qwen or LLaMA GGUF structures via llama.cpp server CUDA containers.

Phase 3: Security, Hardening & Input Sanitization

Mitigating Remote Code Execution hazards from deserializers, enforcing file upload boundaries (e.g. 10MB limits), preventing directory traversal filename exploits, securing CORS configs, and writing robust pytest regression validation structures to secure the endpoints against Prompt DoS and SQL/Key injections.